

Augsburg University

DST 490: Data Visualization for Social Justice Syllabus Spring 2026

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Course Description	General Information	Course Grade	Expectations	Topics Studied
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All course policies are subject to change with notice.

Course Description

From the catalog:

Advanced statistical programming with R to construct data visualization from datasets involving social justice implications. Course content will include regular reflection on ethics and vocation in data science with team and individual applied projects.

The specific learning outcomes of this course are:

- Integrate concepts and skills from previous data science courses to construct engaging and thoughtful data visualizations.
- Apply and critique ethical principles in data science applied to datasets with social justice implications.
- Create a summative body of work representing your data science and visualization skills.
- Reflect on a vocation involving data science and other data-intensive careers.
- Build your advanced data science skill set by:
 - o developing advanced statistical computing skills with R, including functional programming, interactive data graphics, and advanced geospatial analysis.
 - o apply, evaluate, and critique the data science lifecycle (import → tidy → model → visualize)
 - o disseminate your work to multiple audiences (in-person and online).

This course meets the Keystone Core Curriculum requirements

General Information

Credits:	4 credits
Class meets:	Section A: Tuesdays and Thursdays 8:30-10:00 AM in Hagfors 152 Section B: Tuesdays and Thursdays 12:20-1:50 PM in Hagfors 152
Final exam:	Section A: May 5, 2025 from 8:30 - 10:30 AM Section B: May 5, 2026 from 12:00 - 2:00 PM
Professor:	Dr. John Zobitz (zobitz@augsborg.edu)

- Office Hours: I have blocks of time with flexible help hours, offered both in-person at my office (Hagfors 187) or online. You may make an appointment (**in advance**) at the scheduling [link](#).
- Prerequisite: DST 234
- Text & Materials: i) *Modern Data Science with R* (2nd edition, 2021) by Baumer, Kaplan, and Horton - Online version free here: <https://mdsr-book.github.io/mdsr2e/> 🤖
ii) *Data Feminism* (2020) by D'Ignazio & Klein
Online version free here: <https://data-feminism.mitpress.mit.edu/> 🎉
iii) Other open-source readings posted to Moodle.
iv) You also need access to R/RStudio for scientific calculations. Both of these software are free -- see Moodle for information on how to access.

Technology requirements:

You need to be able to complete the following tasks:

- Check email, access Moodle resources including downloading PDFs, and posting PDFs to Moodle.
- Print PDF documents to complete freehand or annotate PDFs electronically.
- Complete data projects using an open source statistical software program and upload assignments to moodle.

Please do not be shy about letting me (or your faculty mentor/academic advisor) know if you need help acquiring or using these resources or if you need help finding a place to work where you can comfortably share video as needed.

More information about access to technology can be found here: [LINK](#)

Course Grade

Your course grade is primarily based on the following categories:

- **Daily Activities (10% of the overall course grade):** Each class session you will engage in activities and exercises that develop your understanding of the topic for the day. You are expected to attend class and engage in the activity for full credit. If you do not attend class (either an excused or unexcused absence), you may receive partial credit for the Daily Activity through completion and submission of an extra assignment related to that day's topic. That assignment will be due at the next class period.
- **Homework (15% of the overall course grade):** Homework is due every day to make sure you keep up and to give you feedback on your progress in mastering course content. The homework will primarily be based on textbook problems (typically analyzed using R). I expect the majority of this homework to be electronically submitted using Quarto. I will provide (re)-instruction in R and Quarto at the start of the term.
- **Ethics & Vocation Journal (15% of the overall course grade):** You will complete a series of readings and assignments ethics and vocation in data science, in response to aspects from *Data Feminism* and subsequent in-class discussion, or from attending and viewing talks that feature data scientists.
- **Quizzes (15% of the overall course grade):** You will have a series of 4 quizzes throughout the term, corresponding to material covered in the course.

- **Data Visualization Portfolio (20% of the overall grade):** You will complete a series of portfolio projects throughout the term that will apply your knowledge and skills in data science using R that also incorporate aspects from the *Data Feminism* textbook. For these portfolio projects, you will create an informative data visualization centered around a particular topic.
- **Data Visualization Project (25% of the overall grade):** Working with a team, you will engage in a data visualization project with an external community partner. This semester-long project will have you regularly contribute to a final product that will be presented at the end of the term. This project will have regular checkpoints throughout the semester, culminating in a final presentation to the community partner at the end of the term.

I reserve the right to limit the amount of work to be submitted late. There will be defined points in the term after which a late submission of an assignment will not be accepted.

At the end of the course, your weighted average is converted to a course grade according to the following scale:

<u>Percentage</u>	<u>Letter</u>	<u>Percentage</u>	<u>Letter</u>	<u>Percentage</u>	<u>Letter</u>
94 – 100%	A	79 – 81.99%	C+	62 – 71.99%	D
92 – 93.99%	A-	74 – 78.99%	C	0 – 61.99%	F
89 – 91.99%	B+	72 – 73.99%	C-		
84 – 88.99%	B	(C- is minimum to pass)			
82 – 83.99%	B-				

Notes:

For this course to count toward the data science major, mathematics major, mathematics minor, computer science minor, or statistics minor, you must earn a grade of C- or higher. For this course to serve as a prerequisite for any MAT course, you must earn a C- or higher. For any other majors or minors not listed here, please consult the respective major or minor description in the Augsburg University course catalog for specific policies.

If you elect to take this course P/LP/NP, then you must earn at least a C- to get grade P. I encourage you to talk with me or with an academic advisor if you are considering the P/LP/N option. The grade of D/LP gives you credit for the course. Under the P/LP/N option, a D grade is recorded as LP and you receive credit for this course, and an F grade is recorded as N and receive no credit for the course.

I reserve the right to lower your course grade if your attendance, punctuality, participation, or preparation falls well below expectations.

If you are not progressing towards a passing grade or if your attendance, punctuality, participation, preparation, or behavior is falling well below expectations, I will usually send a formal Academic Progress Report.

Expectations

This course will deepen and expand your understanding of data science and how you can extend R to create informative and engaging data visualizations and how to disseminate the portfolio of your work in broader contexts. We will integrate your knowledge and skills from previous coursework in the Data Science major.

I want you to be successful in this course. I expect you to give your best effort to meeting these expectations and I will help you learn the material. Here is what it takes to succeed.

- ✓ Buy, rent, or have regular access to course materials.
- ✓ Check your Augsburg email account regularly and know how to access our Moodle site, which includes your course grade.
- ✓ **Attend and participate in every class.** Your learning of data visualization expands when done in community and to provide and receive feedback on your work. Be on time for class, be prepared for the daily activities with a positive attitude, and have all assigned homework completed.

If you must come late or leave early on a special occasion, please let me know in advance. Avoid scheduling appointments that overlap class time. If you have an occasional excused absence, send me an email as soon as possible so that I know you are okay.

- ✓ **Late homework submission policy:** Submit work when originally due. Even if you miss class, work is still due then. While you may submit some late homework, however, no feedback may be provided.

I reserve the right to limit how many and which assignments are eligible for late submission.

I reserve the right to limit the number of make-up assessments and to designate the time period during which make-up assessments are offered.

- ✓ Policies for absences and late work are summarized here: [LINK](#)
- ✓ **Work in team and individual settings.** You will have several opportunities to participate in team and individual projects. For team projects I expect you (with your team members) to have clearly defined roles and responsibilities and support each other throughout the process.
- ✓ **Pay attention during class.** The only thing you should be reading, viewing, writing, or discussing during class is the current material. No texting, taking phone calls, reading, or watching videos during class. Turn your cell phone off or set it to silent. While you may use electronic devices for taking notes (including screenshots of the lecture), participating in class, collaborative whiteboard work, calculations, or approved accommodations, they may not be used for other purposes during class.
- ✓ **Get help when you need it.** I know it is hard to ask for help sometimes, but please feel free to ask me or your classmates. I know the feeling when your code in R doesn't work the way that you intend. Do not wait until the last minute to get help.
- ✓ **Read the course materials.** Read the current material *before* coming to class. Complete the pre-reading. Re-read the material discussed *after* class. Your textbook is a key learning partner and not an index of homework problems. Ask questions on the reading during class or at office hours.
- ✓ Complete all homework assignments on time with an eye on mastery, not the grade. Assigned homework problems are meant to develop your emerging understanding of the

material and emphasize key concepts. (Notice that I did not say you master the concepts so you can do the homework.) Endeavor to do your best on the problems.

- ✓ Form study buddies to work on problems together and discuss concepts. All homework handed in for grading, however, should be your own. Claiming credit for someone else's work is in violation of the Academic Honesty Policy. All calculations on homework problems should be done by hand but you may, of course, check your calculations using technology. If you use technology to solve a problem, make sure you note it accordingly and hand in any printouts with your work.

To be clear, you are encouraged to share ideas, check answers, and compare solutions with classmates, but any work you submit must be “your own work” as denoted by the [Augsburg University Academic Honesty Policy](#).

- ✓ **Statement on Generative AI:** Generative artificial intelligence tools—software that creates new text, images, computer code, audio, video, and other content—have become widely available. Well-known examples include ChatGPT for text and others. This policy governs all such tools, including those released during our semester together. You may use generative AI tools on assignments in this course, within the following limitations.

If you do use generative AI tools on assignments in this class, you must properly document and credit the tools themselves. At the end of the assignment, you are also expected to explain in words how this tool assisted you, and what portion of the work was your own work and what was generated by AI. Because generative AI tools are newer, I will ask you to overdocument your usage as we all collectively work to establish normative use.

If you choose to use generative AI tools, please remember that they are typically trained on limited datasets that may be out of date. Additionally, generative AI datasets are trained on pre-existing material, including copyrighted material; therefore, relying on a generative AI tool may result in plagiarism or copyright violations. Finally, keep in mind that the goal of generative AI tools is to produce content that seems to have been produced by a human, not to produce accurate or reliable content; therefore, relying on a generative AI tool may result in your submission of inaccurate content. It is your responsibility—not the tool's—to assure the quality, integrity, and accuracy of work you submit in any college course.

If you use generative AI tools to complete assignments in this course, in ways that I have not explicitly authorized, I will apply the [Augsburg University Academic Honesty Policy](#) as appropriate to your specific case. In addition, you must be wary of unintentional plagiarism or fabrication of data. Please act with integrity, for the sake of both your personal character and your academic record.

Syllabus statement adapted from here: [LINK](#)

- ✓ As with all Augsburg classes, you should understand and comply with all University policies including:
 - o [Standards of Conduct](#)
 - o [Accessibility Policy](#)
 - o [Discrimination and Bias Policy](#)
 - o [Sexual Misconduct Policy](#)

- ✓ Keep me up-to-date on how the class is going for you. Be sure to set up a meeting if you get an Academic Progress Report so that we can get you back on track.
- ✓ If you need disability-related accommodations to have equal access in this course, please contact the CLASS Office (Disability Resources) at class@augsborg.edu or schedule a meeting with CLASS at www.augsburg.edu/class. If accommodations are required, the CLASS Office will notify me privately about your needs.
- ✓ Creating a classroom environment that welcomes and values all students is very important to me. I ask your assistance in helping support one another by following classroom rules. If you feel unwelcome or unvalued at any time, or have any suggestions for improvement, please let me know so I can try to help. Please do not be shy about telling me what is or is not working for you.
- ✓ Complete any in-class and online course evaluation forms.
- ✓ **Privacy Practices in this Course:** In order to create a classroom community built on trust, our interactions, discussions, and course activities must remain private and free from external intrusion. As a learning community, we are collectively responsible for upholding privacy protections and cultivating inquiry. We respect the individual dignity of all and will refrain from actions that diminish others' ability to learn.
- ✓ **FERPA:** As your instructor, I am committed to protecting your privacy by only using university-approved course technologies and adhering to the Family Educational Rights and Privacy Act ([FERPA](https://www.augsburg.edu/registrar/ferpa/)¹). This includes using your educational data only as allowed by FERPA. I am unable to share any personally identifiable information without your written consent, except for legitimate educational purposes, or under certain conditions described above. FERPA applies to all students enrolled in this institution regardless of sex, race, color, national or ethnic origin, religious belief, age, disability, gender, sexual orientation, gender identity or expression, marital status, familial status, pregnancy, citizenship, creed, genetic information, or veteran status.
- ✓ **Peer to Peer Privacy:** We share a lot with each other in this class, which is why we will prioritize respect, trust, and privacy. Do not share others' personal information, names, phone numbers, emails, addresses, pictures, etc. without that individual's explicit consent. Additionally, all student work should be handled with respect and remain within interactions of this course. Do not photograph, duplicate, digitally or verbally share representations of other students' written or creative work, classroom dialogue, discussion posts, or presentations to social media or any other third party. If you have questions about this please ask me, our department chair, or school director for additional clarification.

¹ <https://www.augsburg.edu/registrar/ferpa/>